# Ratio, Root, and Alternating Series Tests

## Section 1: The Ratio Test

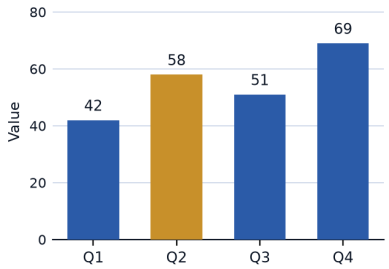
**1** Evaluate the limit $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for the series $\sum_{n=1}^{\infty} \frac{3^n}{n!}$.

**a.** $0$ (converges)

**b.** $3$ (diverges)

**c.** $1$ (inconclusive)

**d.** $\infty$ (diverges)




**2** When is the Ratio Test inconclusive?

**a.** When $L = 1$

**b.** When $L < 1$

**c.** When $L > 1$

**d.** When $L = 0$

**3** Determine the convergence of $\sum_{n=1}^{\infty} \frac{n!}{2^n}$.

**a.** Diverges because $L = \infty > 1$

**b.** Converges because $L = 0 < 1$

**c.** Inconclusive ($L = 1$)

**d.** Converges by Alternating Series Test

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## Section 2: The Alternating Series Test

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**4** What two conditions must an alternating series $\sum (-1)^{n-1} b_n$ with $b_n > 0$ satisfy to converge by the AST?

**a.** $b_{n+1} \le b_n$ for all $n$, and $\lim_{n\to\infty} b_n = 0$

**b.** $b_n$ is increasing, and $\lim_{n\to\infty} b_n = 1$

**c.** $\lim_{n\to\infty} b_n = 0$ only

**d.** The Ratio Test limit must be $< 1$

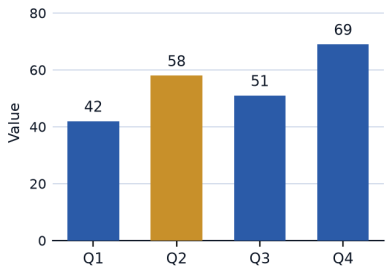
**5** Does the alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ converge or diverge?

**a.** Converges conditionally

**b.** Converges absolutely

**c.** Diverges

**d.** Oscillates without limit




**6** Determine the convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$.

**a.** Converges conditionally

**b.** Converges absolutely

**c.** Diverges

**d.** Inconclusive

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## Section 3: Absolute vs. Conditional Convergence

**7** A series $\sum a_n$ is said to converge absolutely if which condition holds?

**a.** $\sum |a_n|$ converges

**b.** $\sum a_n$ converges but $\sum |a_n|$ diverges

**c.** $\lim_{n\to\infty} a_n = 0$

**d.** All terms are positive

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**8** Which of the following series converges absolutely?

**a.** $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$

**b.** $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

**c.** $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$

**d.** $\sum_{n=1}^{\infty} (-1)^n$

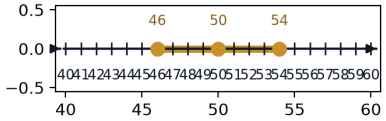
**9** If a series converges absolutely, MUST it also converge in its original form?

**a.** Yes, absolute convergence implies convergence

**b.** No, it may oscillate

**c.** Only if it is alternating

**d.** Only for geometric series




## Section 4: Alternating Series Error Bound

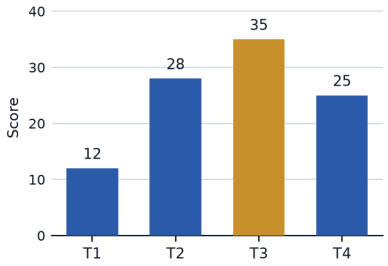
**10** For a convergent alternating series $\sum(-1)^{n-1}b_n$, what is the upper bound for the error $|R_n| = |S - S_n|$?

**a.** $|R_n| \le b_{n+1}$

**b.** $|R_n| \le b_n$

**c.** $|R_n| \le \frac{b_{n+1}}{2}$

**d.** $|R_n| \le \frac{b_n}{n}$




**11** If the sum of $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3}$ is approximated by its first $4$ terms, what is the maximum possible error?

**a.** $\frac{1}{125}$

**b.** $\frac{1}{64}$

**c.** $\frac{1}{216}$

**d.** $\frac{1}{25}$

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**12** How many terms of the alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ are needed to guarantee an error of less than $0.01$?

**a.** $100$ terms

**b.** $99$ terms

**c.** $10$ terms

**d.** $50$ terms

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